

PROGRAMMING FUNDAMENTALS

Spring 2024

29th March

2D Arrays

- *2D arrays*, can hold multiple sets of data
- Think of a two dimensional array as having rows and columns of elements
- Several identical arrays put together
- To store multiple sets of data

Application

- For example, in a grade-averaging program a teacher might record all test scores of one student in an array
- If the teacher has 5 students and 3 tests, that would require 15 variables to store scores of all tests for all 5 students.
- It would be better to define a two-dimensional array
- This 2D array can hold multiple sets of data in rows and columns

2D Array

	Column 0	Column 1	Column 2	Column 3
Row 0	scores[0] [0]	scores[0] [1]	scores[0] [2]	scores[0] [3]
Row 1	scores[1] [0]	scores[1] [1]	scores[1] [2]	scores[1] [3]
Row 2	scores[2] [0]	scores[2] [1]	scores[2] [2]	scores[2] [3]

- The above array has three rows (numbered 0 through 2), and four columns (numbered 0 through 3)
- It has a total of 12 elements in the array

Declaring 2D array

- Two size indices are required
- The first one is for the number of rows
- The second one is for the number of columns

```
int num[3][4];
```

- Declares an integer array with three rows and four columns
- each element has two subscripts: one for its row and another for its column

```
cout << num[0][2];
```

- displays the array element at row 0, column 2:

2D Array

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Row 0	scores[0] [0]	scores[0] [1]	scores[0] [2]	scores[0] [3]
Row 1	scores[1] [0]	scores[1] [1]	scores[1] [2]	scores[1] [3]
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cout << num[0][2];
```

- displays the array element at row 0, column 2:

Declaring 2D array

```
int hours[3][2] = {{8, 5}, {7, 9}, {6, 3}};
```

```
int hours[3][2] = {8, 5, 7, 9, 6, 3};
```

- Both of the above are going to serve the same purpose
- The extra braces that enclose each row's initialization list are optional
- The extra braces visually separate each row

Declaring 2D array

- `int table[3][2] = {{1}, {3, 4}, {5}};`

	c0	c1
r0	1	
r1		
r2		

Declaring 2D array

- `int table[3][2] = {{1}, {3, 4}, {5}};`

	c0	c1
r0	1	
r1	3	4
r2		

Declaring 2D array

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	c0	c1
r0	1	
r1	3	4
r2	5	

Declaring 2D array

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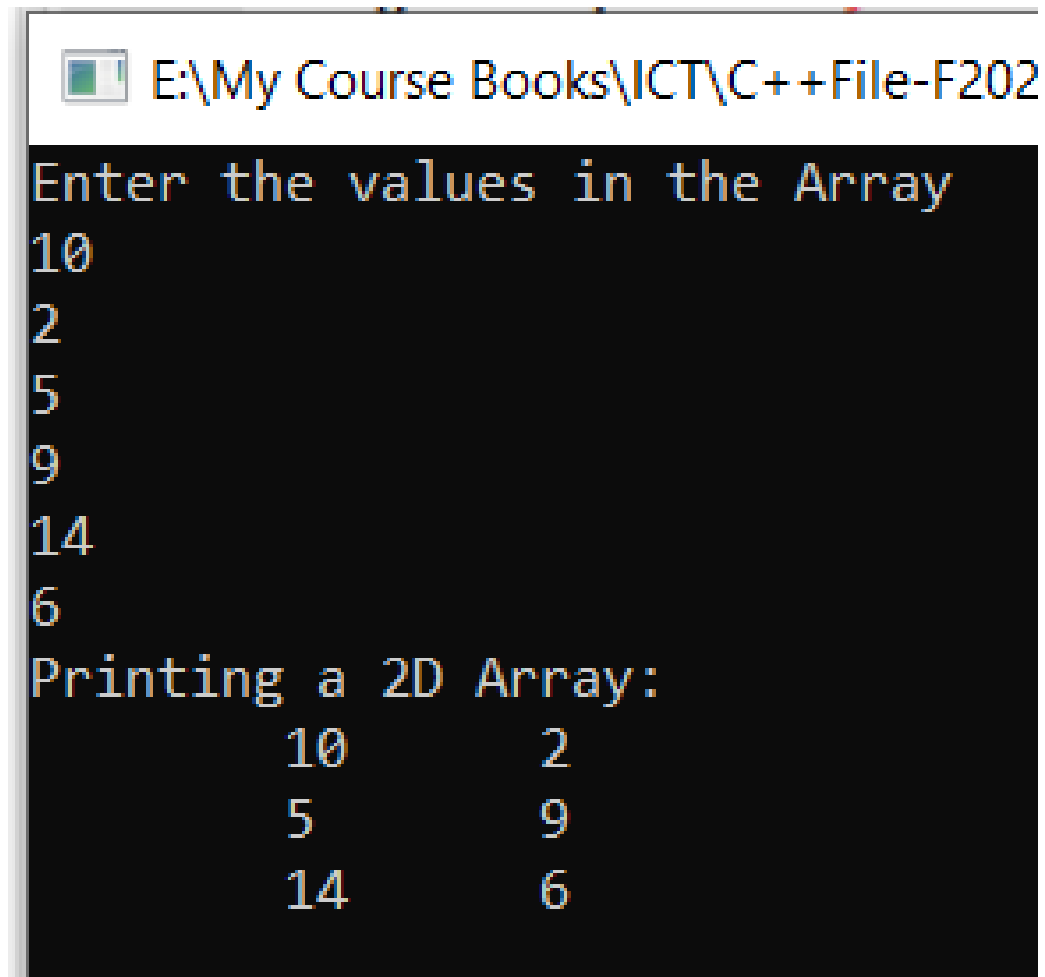
	c0	c1
r0	1	0
r1	3	4
r2	5	0

Ex-1

```
#include <iostream>
using namespace std;
int main()
{
    int hours[3][2];
    cout<<"Enter the values in the Array "<<endl;
    for(int i= 0;i < 3;i++)
    {
        for (int j = 0; j < 2;j++)
        {
            cin>>hours[i][j];
        }
    }
    cout<<"Printing a 2D Array:\n";

    for (int i = 0; i < 3;i++)
    {
        for (int j = 0; j < 2;j++)
        {
            cout<<"\t"<<hours[i][j];
        }
        cout<<endl;
    }
    return 0;
}
```

Output..



The screenshot shows a Windows file explorer window titled "E:\My Course Books\ICT\C++File-F202". The window contains a black terminal window with the following text:

```
Enter the values in the Array
10
2
5
9
14
6
Printing a 2D Array:
      10      2
      5      9
      14      6
```

10	2
5	9
14	6

Ex-2

```
#include <iostream>
using namespace std;
int main()
{
    int hours[3][2];
    cout<<"Enter the values in the Array "<<endl;
    int sum = 0;
    for(int i= 0;i < 3;i++)
    {
        for (int j = 0; j < 2;j++)
        {
            cin>>hours[i][j];
            sum = sum + hours[i][j];
        }
    }
    cout<<"Printing a 2D Array:\n";

    for (int i = 0; i < 3;i++)
    {
        for (int j = 0; j < 2;j++)
        {
            cout<<"\t"<<hours[i][j];
        }
        cout<<endl;
    }
    cout<<endl<<"Sum of all values in the Array = "<<sum;
    return 0;
}
```

Output..



E:\My Course Books\ICT\C++File-F2023\2024\Lec

```
Enter the values in the Array
```

```
1
```

```
2
```

```
3
```

```
4
```

```
5
```

```
6
```

```
Printing a 2D Array:
```

```
1
```

```
2
```

```
3
```

```
4
```

```
5
```

```
6
```

```
Sum of all values in the Array = 21
```

```
-----
```


Book Reference

- D.S. Malik (ch - 8)
- Tony Gaddis (ch - 7)